John Paul Helveston, Ph.D.

Engineering Management and Systems Engineering
The George Washington University
Science & Engineering Hall, Office 2830
800 22nd St NW, Washington, DC 20052
☎ +1 (202) 994-7173
🖂 jph@gwu.edu
www.jhelvy.com

Academic Appointments

Visiting Scholar — *Dhaka, Bangladesh*
Department of Public Health, Daffodil International University — Sep 2025 - Present

- Collaborate directly with the Head of the Department of Public Health to strategically enhance the department's research capacity and infrastructure.

Visiting Scholar — *Dhaka, Bangladesh*
Department of Public Health, Daffodil International University — NA - NULL

- Mentor students in designing research projects and drafting manuscripts targetted for high-impact, ISI and Scopus-indexed (Q1/Q2) journals.
- Co-author and develop competitive research proposals to secure local and international external funding sources in partnership with DIU researchers.
- Participate in institutional research progress workshops and deliver specialized online courses as assigned by the department.

Faculty (Bioinformatics) — *Dhaka, Bangladesh*
cBLAST, University of Dhaka — Aug 2023 - Present

- Teach specialized modules including Biomedical Machine Learning with Python and Data Science and Machine Learning for Biologists.

Faculty (Bioinformatics) — *Dhaka, Bangladesh*
cBLAST, University of Dhaka — NA - NULL

- Train students to leverage High-Performance Computing (HPC) infrastructure for scaling bioinformatics, AI models and handling complex biological datasets.
- Provide technical consultation and hands-on data analysis support to cBLAST students executing computational biology workflows.
- Collaborate with Dr. Zeba Islam Seraj to deliver advanced computational training that effectively bridges biological data with machine learning frameworks.

Program Lead — *Dhaka, Bangladesh*
GSA Bioinformatics Internship — Jul 2025 - Present

- Lead a multi-institutional bioinformatics capacity-building initiative in partnership with GNOBB, ASI School of Life, and SPSB to bridge the gap between formal education and reproducible research workflows

Program Lead *Dhaka, Bangladesh*
GSA Bioinformatics Internship NA - NULL

- Design and execute a structured curriculum for student cohorts, successfully graduating Cohort 01 and currently directing the training for Cohort 02.
- Train and mentor multiple cohorts of interns in standardized Single-Cell RNA-Seq and multi-omics pipelines via the GNOBB-ASI Bioinformatics Hub.
- Manage the cross-institutional logistics and program delivery from Dhaka, expanding access to hands-on computational biology mentorship across Bangladesh.

Program Lead *Dhaka, Bangladesh*
Bioinformatics Mentorship Program Apr 2026 - Present

- Design and direct a 12-week structured capacity-building curriculum focused on standardizing reproducible RNA-Seq research across dedicated cancer genomics tracks.

Program Lead *Dhaka, Bangladesh*
Bioinformatics Mentorship Program NA - NULL

- Manage competitive student intake and program delivery, successfully graduating Cohort 01 and currently leading active mentorship for Cohort 02.
- Embed open-science practices from day one, training 10 mentees per cohort to utilize version-controlled workflows and reproducible methodologies.
- Guide participants through the end-to-end research lifecycle, transforming raw computational analyses into manuscript-ready outputs targeted for peer-reviewed publication.

Teaching Assistant *Dhaka, Bangladesh*
Department of Microbiology, Jagannath University Oct 2022 - 1669852800

- Deliver targeted research methodology training to over 40 undergraduate students, standardizing experimental design and statistical analysis practices within public health microbiology.

Teaching Assistant *Dhaka, Bangladesh*
Department of Microbiology, Jagannath University NA - NULL

- Support departmental infectious disease surveillance data processing by managing and verifying incoming laboratory and epidemiological datasets.
- Critically review student research proposals and literature assessments to cultivate rigorous scientific reasoning and adherence to academic writing standards.
- Assist senior faculty during laboratory practicals, ensuring undergraduate students master sterile techniques, media preparation, and microbial identification protocols.

Education & Training

Master of Science in Microbiology *Dhaka, Bangladesh*
Jagannath University 2019—2020

- Medium of Instruction: English
- Capstone Project: Genomic surveillance, clinical epidemiology, and outbreak response frameworks for Chikungunya in Bangladesh
- Supervisor: Dr. Syeda Tasneem Towhid

Bachelor of Science in Microbiology *Dhaka, Bangladesh*
Jagannath University 2016—2019

- Medium of Instruction: English
- Capstone Project: Benchmarking Whole Genome Sequencing (WGS) Data Analysis Pipelines: A State-of-the-Art Performance Review in Microbial Genomics
- Supervisor: Dr. Syeda Tasneem Towhid

Certificate Program in Interdisciplinary Computational Biology *Dhaka, Bangladesh*
BRAC University 2026

- Completed an intensive training curriculum under the Certificate Program in Interdisciplinary Computational Biology at BRAC University, specializing in cross-disciplinary data workflows.
- Master computational frameworks and statistical tools required to analyze complex biological systems and bridge traditional life science methodologies with computer science.

Specialized Training in AI for Public Health *Dhaka, Bangladesh*
Child Health Research Foundation (CHRF) 2023

- Complete an intensive, specialized training curriculum focused on the intersection of artificial intelligence and epidemiology at the Child Health Research Foundation (CHRF).
- Master the application of machine learning algorithms and predictive modeling to address pressing challenges in public health and infectious disease surveillance.

Research Interests

- **Clean Technology Development & Adoption**: Quantify how consumers, firms, markets, and policy affect the nature & pace of transitioning to sustainable energy and transportation technologies.
- **Consumer Preferences for Emerging Technology**: Measure and model consumer preferences to assess policy and product design and simulate consumer choice behavior under uncertainty.
- **Electric Vehicles & Sustainable Transportation Technologies**: Assess barriers and opportunities to accelerating the development & adoption of sustainable transportation technologies.
- **U.S.-China Climate Relationship**: Study the critical relationship between the US and China in developing, scaling, and mass deploying low carbon energy technologies.

Teaching Interests

- **Programming & Data Analytics**: Programming in R and Python; exploratory data analysis; data visualization; reproducibility.
- **Choice Modeling**: Discrete choice modeling; consumer preferences and choice behavior; conjoint analysis; design decisions.
- **Team Projects**: Open-ended, team-based projects that emphasize critical thinking and real-world data collection and analysis.

External Grants

Total External Funding: $1,676,128

Total External Funding as PI: $1,246,128

A. Principal Investigator

1. Alfred P. Sloan Foundation Energy & Environment Program. "Quantifying the Benefits and Constraints of Plug-In Electric Vehicle Smart Charging Adoption". With Brian Tarroja (UCI, Co-PI), Eric Hittinger (RIT), Alan Jenn (UC Davis). $516,677. Helveston portion $282,571. Sep. 2023 - Aug. 2026.
2. DOE Vehicle Technologies Office (VTO). "Quantifying New and Used Plug-in Electric Vehicle Market Dynamics in Disadvantaged Communities". With Dong-Yeon (D-Y) Lee (NREL), Jeff Gonder (NREL). $474,517. Helveston portion $276,838. Oct. 2023 - Jun. 2026.
3. Electric Power Research Institute. "Investigation of used electric vehicle movements, sales and availability". $109,743. 100% Credit. Jul. 2022 - Jun. 2023.
4. Alfred P. Sloan Foundation Energy & Environment Program. "Identifying More Efficient and Equitable Plug-In Electric Vehicle Financial Incentives Through Consumer-Centric Design". With Laura Roberson. $31,268. 100% Credit. Sep. 2021 - Aug. 2022.
5. Toyota Mobility Foundation. "Evolvability Analysis Framework Dashboard Development". With Zoe Szajnfarber (Co-PI) and David Broniatowski (Co-PI). $24,963. 100% Credit. Jul. 2021 - Jun. 2022.
6. New Venture Fund Public Interest Technology University Network. "GW Coders Scholarship & Internship Program: Building Coding Capacity for Public Interest Technology Engagement". With Ryan Watkinds (Co-PI). $88,960. 100% Credit. Jul. 2021 - Jun. 2022.

B. Co-Principal Investigator

1. Toyota Mobility Foundation. "Evolvability for Mobility Systems". With Zoe Szajnfarber (PI) and David Broniatowski (Co-PI). $430,000. Helveston portion $238,150. Jul. 2019 - Jun. 2021.

Internal Grants (GWU)

1. REACH Center Seed Research Grant Program Proposal. "Quantifying the Air Quality and Environmental Justice Implications of Disparities in Access to Clean Vehicles". With Gaige Kerr. $40,000. 100% Credit. Mar. 2025 - Apr. 2026.
2. Sustainability Alliance Seed Grant, Energy Technology and Decision Making Research Cluster. "Analyzing the Spatial Dimensions of Consumer Buying Patterns for Clean Vehicle Technologies". $3,500. 100% Credit. Jun. 2025 - Aug. 2025.
3. Adapting Course Materials for Equity, GW Libraries & Academic Innovation (LAI). "Textbook for Open Source Course: Exploratory Data Analysis". $1,500. 100% Credit. Jul. 2024 - Jun. 2025.
4. GWU University Facilitating Fund. "Measuring Consumer Preferences for Used Alternative Fuel Vehicles". $14,210. 100% Credit. Jul. 2021 - Jun. 2022.
5. GWU University Facilitating Fund. "Spatial and Temporal Mapping of Technological Progress In Electric Vehicle Powertrain Technologies". $16,929. 100% Credit. Jul. 2019 - Jun. 2020.

Honors / Awards

- 2024: Best paper award, Bridging Transportation Researchers (BTR) #6 Conference.
- 2024: Representative for Shanghai Workshop on Climate and China-U.S. Relations, organized by Fudan-UCSD Center on Contemporary China (Shanghai, China).
- 2023: SEAS Outstanding Early Career Teaching Award.
- 2023: IOP Outstanding Reviewer Award.

- 2022: Awarded an "IOP Trusted Reviewer" status for a 5/5 rated review for *Environmental Research Letters*.
- 2022: Best poster award, Integrated Assessment Modeling Consortium.
- 2019: Energy Innovation Policy and Management Scholar, Innovation and Information Technology Foundation.
- 2018: Finalist, Coase Dissertation Award, Society for Institutional & Organizational Economics.
- 2017: Best Dissertation Award, Industry Studies Association.
- 2016: Best Paper in Innovation & Entrepreneurship Research, Industry Studies Association.
- 2014: NSF East Asia Pacific Summer Institute Fellowship ($5,000).
- 2014: Link Energy Foundation Fellowship.
- 2013: Herbert Toor Award for Best Engineering and Public Policy Qualifying Examination Paper.
- 2011: NSEP Boren Fellowship (Award Declined).
- 2010: Taiwan Huayu Mandarin Enrichment Scholarship.
- 2009: Department of State Critical Language Scholarship for Mandarin Chinese.
- 2007: Horton Honors Scholarship for 6-month independent study abroad in China.
- 2005: Eleanor Davenport Leadership Scholarship, Virginia Tech: Full tuition and Fees (2005 - 2009).
- 2002: Eagle Scout Award, BSA Troop 11, Valrico, FL, July 7, 2002.

Publications

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ORCID: 0000-0002-2657-9191 | Google Scholar Profile

Underline indicates advisee; * graduate student, **undergraduate student

Note on author order: In my field, it is conventional that the last author denotes intellectual direction by the senior author when advised student(s) are engaged on the study. On non–student papers, authorship effort is denoted in standard order.

A. Refereed Journal Articles

1. **Hossain, M. J.**, Towhid, S.T., Sultana, S., Mukta, S.A., Gulshan, R., Miah, M.S. (2022) "Knowledge and Attitudes towards Thalassemia among Public University Students" *Microbial Bioactives*. 5(2). 10.25163/microbbioacts.526325 Survey-based study on thalassemia awareness among the Bangladeshi youth.
2. Towhid, S.T., **Hossain, M. J.**, Sammo, M.A.S., & Akter, S. (2022) "Perception of Students on Antibiotic Resistance and Prevention" *Eur. J. Bio. Biotech*. 3(3). 10.24018/ejbio.2022.3.3.341 Investigates student awareness and attitudes toward the antibiotic resistance crisis.
3. ***Hossain, M. J.**, Islam, M. W., Munni, U. R., Gulshan, R., Mukta, S. A., Miah, M. S., Sultana, S., Karmakar, M., Ferdous, J., & Islam, M. A. (2023) "Health-related quality of life among thalassemia patients in Bangladesh (SF-36)" *Scientific Reports*. 13(1). 10.1038/s41598-023-34205-9 Assesses the physical and mental health burden on patients living with thalassemia.
4. Akter, M. M., & ***Hossain, M. J.** (2024) "Food consumption patterns and sedentary behaviors among university students" *Health Science Reports*. 7, e70259. 10.1002/hsr2.70259 Links dietary habits and lack of activity to health risks in students.
5. ***Hossain, M. J.**, Azad, A. K., Shahid, M. S. B., Shahjahan, M., & Ferdous, J. (2024) "Prevalence, antibiotic resistance pattern for bacteriuria from patients with UTIs" *Health Science Reports*. 7, e2039. 10.1002/hsr2.2039 Maps antibiotic resistance landscapes for UTI-causing bacteria in clinical settings.

6. *__Hossain, M. J.__, Das, M., Islam, M. W., Shahjahan, M., & Ferdous, J. (2024) "Community engagement and social participation in dengue prevention" *Health Science Reports*. 7, e2022. 10.1002/hsr2.2022 Evaluates how social participation impacts the effectiveness of dengue control.
7. Islam, M. W., Shahjahan, M., Azad, A. K., & *__Hossain, M. J.__ (2024) "Factors contributing to antibiotic misuse among parents in Dhaka City" *Scientific Reports*. 14, 2318. 10.1038/s41598-024-52313-y Identifies parental socio-economic factors driving antibiotic misuse in children.
8. *__Hossain, M. J.__, Das, M., & Munni, U. R. (2024) "Urgent call for compulsory premarital screening for thalassemia prevention in Bangladesh" *Orphanet Journal of Rare Diseases*. 19, 326. 10.1186/s13023-024-03344-1 Advocates for policy-driven premarital screening to reduce thalassemia incidence.
9. *__Hossain, M. J.__, Sony, S. A., Fariha, F. T. J., & Hossen, S. (2025) "Preventing the silent threat: A perspective on preparing Bangladesh for Human Metapneumovirus" *Health Science Reports*. 8, e71101. 10.1002/hsr2.71101 Strategic perspective on preparing health systems for HMPV outbreaks in Bangladesh.
10. Das, M., & **Hossain, M. J.** (2025) "Young stroke in Bangladesh: Addressing rare cases with diagnostic challenges" *Stroke and Vascular Neurology*. svn-2025-004178. 10.1136/svn-2025-004178 Call for action on diagnosing and managing rare stroke cases in young Bangladeshi patients.
11. *__Hossain, M. J.__, Das, M., Shahjahan, M., Islam, M. W., & Towhid, S. T. (2025) "Clinical and hematological manifestation of dengue patients in 2022 outbreak" *Health Science Reports*. 8, e70356. 10.1002/hsr2.70356 Study details clinical profiles and blood changes during a major Dhaka dengue outbreak.
12. Shanta, A. S., Islam, N., Al Asad, M., Akter, K., Habib, M. B., **Hossain, M. J.**, Nahar, S., Godman, B., & Islam, S. (2024) "Resistance and co-resistance of metallo-beta-lactamase genes in diarrheal and urinary-tract pathogens" *Microorganisms*. 12(8), 1589. 10.3390/microorganisms12081589 High prevalence of MBL genes found in clinical pathogens within Bangladesh.
13. Bari, S. M., Fuad, M., **Hossain, M. J.**, Ahammad, I., Begum, M., & Helal, M. M. U. (2025) "A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout" *BMC Genomics*. 26, 999. 10.1186/s12864-025-12127-2 Meta-analysis reveals conserved transcriptomic signatures of stress in rainbow trout.
14. Ahmed, M. Z., Billah, M. M., Ferdous, J., & *__Hossain, M. J.__ (2025) "Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors" *Scientific Reports*. 15, 14405. 10.1038/s41598-025-88339-z CCT5 identified as a pan-cancer biomarker linked to immune infiltration and prognosis.
15. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & *__Hossain, M. J.__ (2025) "Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer" *Scientific Reports*. 15, 34490. 10.1038/s41598-025-13315-6 WNT genes serve as critical prognostic biomarkers and immune regulators in breast cancer.

B. Refereed Articles in Conference Proceedings

C. Working Papers & Papers Under Review

1. **Hossain, M. J.**, Dev, P.C. (2026) "Fezf2-mediated cortical development: Multi-omics analysis" *Working Paper*. https://github.com/hossainlab/fezf2-multiomics
2. **Hossain, M. J.**, Dev, P.C. (NA) "EWSAtlas: A Harmonized Single-Cell Transcriptomics Atlas of Human Ewing sarcoma" https://github.com/hossainlab/EWSAtlas
3. **Hossain, M. J.**, Dev, P.C., & Shahariar, M. (2026) "Deep Learning-Driven AMR Drug Repurposing" *Working Paper*. https://github.com/hossainlab/amr_drug_repurposing
4. **Hossain, M. J.** (2026) "DeepAMR: AI-powered antimicrobial resistance prediction from genomic data" *Working Paper*. https://github.com/hossainlab/DeepAMR

5. **Hossain, M. J.**, **Tabassum, R., **Ahmad, F., **Ahamed, M. S., **Aditee, L. M., Fuad, M., Shahjahan, M. (2025) "The Pan-Cancer Consensus Transcriptome (PCCT): A Multi-Layered Meta-Analysis and Machine Learning Framework for Cross-Platform Biomarker Discovery" *Working Paper*. https://github.com/gsabioinfointernship/Pancreatic_Cancer_Meta_Analysis/

D. Books and Book Chapters

E. Magazine Publications

F. Opinion Editorials

G. Podcasts

H. Other Publications

I. Theses

J. Software

Presentations / Conferences

Total Presentations by Category:

`Invited Speaker (4), Conference Presentation (3), \newline Poster Presentation (4)`

A. Invited Speaker

1. "Undergraduate Research - Importance, Benefits, and Challenges". CHIRAL Bangladesh. Dhaka, Bangladesh. Mar 21, 2019.
2. "State the Art of Microbial Genome Analysis". Jagannath University. Dhaka, Bangladesh. May 21, 2018.
3. "Computational Biology and Bioinformatics Research in Resource-Limited Settings: Strategies, Tools and Opportunities". Jagannath University Higher Study and Research Society. Dhaka, Bangladesh. Jun 25, 2015.
4. "Mastering Biomedical Data Management". IFMSA Bangladesh. Dhaka, Bangladesh. Apr 21, 2014.

B. Invited Panelist

C. Conference Presentations

1. "Fezf2-Mediated Cortical Development: Multi-Omics Single-Cell Analysis", with *Islam, MM. Network of Young Biotechnologists Bangladesh. Dhaka, Bangladesh. Apr 20, 2026.
2. "Diagnostic and Electrophysiological Features of Hirayama Disease in Young Adult Male: A Case Report", with Manisha Das. Institute of Epidemiology, Disease Control and Research (IEDCR). Dhaka, Bangladesh. May 21, 2020.

3. "Quantitative Microbial Risk Assessment from Vancomycin-Resistant Enterococcus faecalis and Enterococcus faecium from a Specific Neighborhood in Dhaka City, Bangladesh", with Nayeem, M.U., Mrittika, M.A., Azad, A.K., Ferdous, J., Ahmed, S., Sanyal, S.K., Towhid, S.T.. Bangladesh Society of Microbiologists. Dhaka, Bangladesh. May 21, 2019.

D. Conference Panel Organizer / Chair

E. Posters

1. "Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis", with **Ahmad, F., Fuad, M., Shahjahan, M., & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
2. "Network-Driven Bioinformatics Approach Uncovers Prognostic Signatures and Therapeutic Targets in Pancreatic Ductal Adenocarcinoma", with **Aditee, L. M., Fuad, M., Shahjahan, M., & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
3. "Bioinformatic Analysis of Transcriptomic Data Reveals Molecular Signatures Driving Colorectal Cancer", with **Ahamed, M. S., Fuad, M., Shahjahan, M., & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
4. "Comparative Transcriptomic Meta-Analysis Reveals Shared and Distinct Transcriptional Networks and Pathway Crosstalk in Hepatocellular Carcinoma for Biomarker and Therapeutic Target Identification", with **Tabassum, R., Fuad, M., Shahjahan, M., & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.

F. Discussant

Selected Media Coverage: Impact of Research in Society

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Total Media Appearances

National Press (2)

1. May 27, 2024, **The Business Standard**: Quoted in article What the rise in self-medication tells us about the country's healthcare system.
2. Feb 08, 2024, **The Business Standard**: Study featured in article "Parental lack of antibiotic knowledge imperils child health in Bangladesh: Study".

Teaching & Education

A. Courses Taught at George Washington University

B. Workshops

1. "Publication-ready Tables with R". CHIRAL Bangladesh. Dhaka, Bangladesh. Apr 07, 2025.
2. "Building Dashboard with R". CHIRAL Bangladesh. Dhaka, Bangladesh. Apr 07, 2025.
3. "Easystats for Biomedical Researchers". CHIRAL Bangladesh. Dhaka, Bangladesh. Apr 07, 2025.

Sem.	Course	Level	N. Resp. / N. Enrolled	Instr. FCE∗ / Dept Mean
C01	AI4DD: AI in Drug Discovery: In Silico Toxicology Modeling	Ugrad	13 / 18	5 / 4.3
C02	AI4LS: AI for Life Sciences	Mixed	16 / 23	5 / 4.3
C04	bulkRNASeq101: Bulk RNA-Seq Analysis for Absolute Beginners	Ugrad	9 / 18	4.8 / 4.4
C03	scRNASeq101: Single-Cell for Absolute Beginners	Ugrad	15 / 21	4.9 / NA
C01	CCG101: Computational Cancer Genomics	Ugrad	13 / 22	5 / 4.4
C01	NGS101: NGS Data Analysis for Absolute Beginners	Mixed	18 / 21	4.9 / 4.4
C01	CN101: Computational Neurogenomics	Mixed	24 / 32	4.9 / 4.4

* Faculty Course Evaluations (FCE) are scored by students (1 = worst, 5 = best).

† No course evaluations were recorded in Spring 2020 due to the COVID19 pandemic.

4. "R for Bioinformatics". CHIRAL Bangladesh. Dhaka, Bangladesh. Apr 07, 2025.
5. "Data Analysis with R". CHIRAL Bangladesh. Dhaka, Bangladesh. Apr 07, 2025.
6. "R for Cancer Bioinformatics". CHIRAL Bangladesh. Dhaka, Bangladesh. Apr 07, 2025.
7. "R for Research". CHIRAL Bangladesh. Dhaka, Bangladesh. Oct 01, 2023.
8. "Machine Learning for Bioinformatics". CHIRAL Bangladesh. Dhaka, Bangladesh. Aug 01, 2023.
9. "RNA-Seq Analysis with R". CHIRAL Bangladesh. Dhaka, Bangladesh. Aug 01, 2023.
10. "AI for Drug Discovery". CHIRAL Bangladesh. Dhaka, Bangladesh. Aug 01, 2023.
11. "Cancer Bioinformatics". CHIRAL Bangladesh. Dhaka, Bangladesh. Aug 01, 2023.
12. "Python for Health Data Analytics". CHIRAL Bangladesh. Dhaka, Bangladesh. Aug 01, 2023.
13. "Statistical Analysis with R for Research". DeepBio Academy, CHIRAL Bangladesh. Dhaka, Bangladesh. Oct 01, 2014.
14. "Statistical Analysis with R for Research". Genetic Engineering Club, JnU. Dhaka, Bangladesh. Oct 01, 2014.

C. Seminars

D. Guest Lectures

E. Educational Contributions

1. Courses Developed at GWU

- **EMSE 4571 / 6571: Intro to Programming for Analytics**. Developed new open-source introductory level course that provides a foundation in programming for data analytics using the R programming language with a comparison to Python.
- **EMSE 4572 / 6572: Exploratory Data Analysis**. Developed new open-source course that provides a foundation in exploring data using the R programming language, including how to source, manage, wrangle, explore, and visualize a wide variety of data types. All analyses are reproducible from raw data to results using RMarkdown and Quarto. Students demonstrate mastery of these skills through a semester-long research project of their own design, culminating in a reproducible final report and a 10-minute presentation of their findings.
- **EMSE 6035: Marketing Analytics for Design Decisions**. Developed a new open-source course that introduces data analysis techniques to inform design decisions in an uncertain, competitive market. Over the course of the semester, students learn and apply theory and methods to a team project to assess the market competitiveness of an emerging product or technology. Students learn how to design and field conjoint surveys as well as how to source, manage, and visualize data and modeling results using the R programming language. Students demonstrate mastery of

these skills through a semester-long research project of their own design, culminating in a final report and a 10-minute presentation of their design insights.

2. Resources Developed

- **Textbook**: "Programming for Analytics in R", written for *EMSE 4571 / 6571: Intro to Programming for Analytics*.
- **Textbook**: "Yet Another R Dataviz (YARD) Book", written for *EMSE 4572 / 6572: Exploratory Data Analysis*.
- **Video lecture series** for *EMSE 6035: Marketing of Technology*. Six-lecture series on conjoint survey theory and practice. View on youtube.
- **Autograder** for *EMSE 4571 / 6571: Intro to Programming for Analytics*. Students can test and receive automated feedback on their programming assignments.
- **Course website**: http://p4a.seas.gwu.edu/, for *EMSE 4571 / 6571: Intro to Programming for Analytics*. Open source lessons on the fundamentals of programming for data analytics in R with a comparison to Python.
- **Course website**: http://eda.seas.gwu.edu/, for *EMSE 4572 / 6572: Exploratory Data Analysis*. Open source lessons on sourcing, managing, transforming, and exploring a wide variety of data types in R.
- **Course website**: http://madd.seas.gwu.edu/, for *EMSE 6035: Marketing Analytics for Design Decisions*. Open source lessons on designing conjoint surveys and choice modeling in R.

Advising

A. Primary Research Advisor

1. Ph.D. Students - Current

2. Ph.D. Students - Completed

3. Masters Students - Completed

4. Undergraduate Students - Current

1. Md. Shakil Ahamed, Mawlana Bhashani Science and Technology University (2025 to Present).
2. Rahnuma Tabassum, Jagannath University (2025 to Present).
3. Fayez Ahmad, Notre Dame College, Dhaka (2025 to Present).
4. Sabbir Khan, Gazipur Agricultural University (2025 to Present).
5. Lamisa Manha Aditee, BRAC University (2025 to Present).

5. Undergraduate Students - Completed

4. High School Students

B. Ph.D. Committee Member

1. Ph.D. Students - Current

2. Ph.D. Students - Completed

Awards / Honors by Advised Students

- 2024: Winner of the GW OSPO Open Source Project Award.
- 2024: PhD student Leah Kaplan awarded First Prize for GWU 3 Minute Thesis competition.
- 2023: PhD student Leah Kaplan selected as FASPE Design & Technology Fellow.
- 2023: EMSE Senior design team with Eliese Ottinger wins best paper at the IEEE Systems and Information Engineering Design Symposium (SIEDS).
- 2022: PhD student Laura Roberson awarded First Prize for Studies in Engineering and Data Analysis at the GW Research Showcase.
- 2022: PhD student Laura Roberson awarded Sustainability and Resiliency Prize at the GW Research Showcase.
- 2020: PhD student Leah Kaplan awarded NSF Graduate Student Research Fellowship ($147,000).

Service

A. Departmental Service (EMSE)

1. Non-Academic Short Course Committee (2024 - present).
2. Undergraduate ABET Committee (2024 - present).
3. Undergraduate Curriculum Committee (2020 - present).
4. Advised a senior design team (2022 – 2023).
5. Scribe for Faculty Meetings (2019 - 2022).
6. Doctoral Qualifying Exam Committee (2019 - 2020).

B. College Level Service (School of Engineering and Applied Sciences, GWU)

1. *Director*, Data Analytics MS Program (2023 - present).
2. Strategic Research Planning Committee (2021 - 2022).
3. Computing Committee (2020 – 2022, 2024 - present).
4. Community Development: *Co-founder* and bi-weekly research meet up community organizer, GW Coders Organization (2020 - present).

C. University Level Service (George Washington University)

1. *Event Organizer*, Electric Vehicle Policy Symposium, GW Alliance for a Sustainable Future (2025).
2. *Workshop Organizer*, Coding with AI (full-day workshop for students and faculty) (2023).
3. Generative AI Faculty Committee (2023 - present).

D. Professional Service

1. Organized research symposium for the International Electric Vehicle Research Council held in SEH (2024, 2025).
2. *Special Issue Editor*, Environmental Research Infrastructure and Sustainability (2024 - present).
3. Annual Conference Organizing Committee Member, Industry Studies Association (2023 - present).
4. Dissertation Award Committee Chair, Industry Studies Association (2020 - 2022).
5. Online Program Committee Member, Industry Studies Association (2020 - 2022).
6. *Organizer*, Industrial Policy Webinar Series, Industry Studies Association (2021 - 2022).
7. Dissertation Award Committee Member, Industry Studies Association (2018 & 2019).
8. Conference Organizing Committee Member, Technology, Management, & Policy Consortium (2019).
9. Review Panel Member, National Science Foundation (2021).
10. Review Panel Member, VTO Annual Merit, Dept. of Energy (2020 & 2021).
11. IEEE Vehicular Power and Propulsion Conference Reviewer (2020 & 2024).
12. Reviewer, National Academies Transportation Research Board Annual Meeting (2013, 2019, & 2020).

E. Peer Reviewer

Completed 85 journal reviews and 14 conference paper reviews (2013 - present)

List of Journals

Climate Policy

Energies
Energy Policy
Environmental Research Letters
Environmental Research: Infrastructure & Sustainability
Geography & Sustainability
iScience
Joule
Journal of Environmental Management
Journal of International Business
Journal of Mechanical Design
Journal of Systems Science & Systems Engineering
Nature Communications Earth & Environment

Nature Energy

Research Policy
RSC Sustainability
Science
Science Advances
Technovation
The Electricity Journal
Transactions in GIS
Transactions on Engineering Management
Transport Policy
Transportation Research Part A: Policy & Practice
Transportation Research Part D: Transport & Environment
World Electric Vehicle Journal

Memberships in Professional Organizations

- Transportation Research Board
- Industry Studies Association
- Academy of Management
- INFORMS
- U.S. Association for Energy Economics (USAEE / IAEE)
- Tau Beta Pi
- Phi Beta Kappa
- Beijing Energy Network

Industry Experience

- Intern, Electric Vehicle Charging Policy. Innovation Center for Energy & Transportation (iCET), Beijing, China (Jan. - May., 2011).
- Engineering Intern, Wind Power Advanced Technology Operations. General Electric Company, Shanghai, China (Aug. - Dec., 2008).
- Engineering Intern, Wind Power Advanced Technology Operations. General Electric Company, Greenville, SC, USA (Jun. - Aug., 2007).

Leadership, Volunteer, and Community Activities

- *Co-Founder and Organizer*, GW Coders: Informal study group to apply computational and data analytics skills in research (2020-present).
- *Analyst & Committee Member*, Boston University Climate Action Plan Task Force (2016 - 2018).
- *Violinist*, Carnegie Mellon All University Orchestra, 1st Violin Section (2011 - 2015). _ Dance Instructor_, Tartan Swing (CMU Swing Dance Club) (2011 - 2015).
- *Head Dance Instructor*, Solely Swing, Virginia Tech Swing Dance Club (2007 - 2010).
- *Concert Master*, New River Valley Symphony Orchestra, Virginia Tech (2005 - 2010).
- *Volunteer*, Virginia Tech Alternative Breaks Service Programs for Tau Beta Pi, Appalachia Service Project, & Presbyterian Campus Ministries (2006 - 2009).

Skills

- **Language**: Mandarin Chinese (speaking: *fluent*, reading / writing: *intermediate*).
- **Modeling & Analysis**: Discrete Choice Modeling, Conjoint Analysis, Decision Analysis, Monte Carlo Simulation, Consumer Preferences, Quantitative Policy Analysis, Process-Based Cost Modeling, Optimization, Regression.
- **Data Collection**: Survey Design, Qualitative Interviews.
- **Programming**: R, Python, Git, MatLab, LaTeX, Shiny, Stata, Mathematica, HTML, Wordpress.
- **Software**: Adobe Photoshop, Adobe Illustrator, Microsoft Office, Analytica.

Dance Awards

A. Lindy Hop

- **1st Place**: 2016 Advanced Jack & Jill w/Banban, *China Lindy Hop Championships*, Beijing, China.
- **3rd Place**: 2013 Open Jack & Jill, *Rocktober*, Columbus, OH.
- **5th Place**: 2012 Open Jack & Jill, *Boston Tea Party*, Boston, MA.
- **2nd Place**: 2012 Open Jack & Jill w/Akemi Kinukawa, *Babble*, New York, NY.
- **1st Place**: 2011 Open Strictly Lindy w/Annabel Truesdell Quisao, *International Lindy Hop Championships*, Washington, D.C.
- **Finals**: 2011 Open Jack & Jill, *Lindy Focus X*, Asheville, NC.
- **Finals**: 2011 Open Jack & Jill, *International Lindy Hop Championships*, Washington, D.C.

B. Solo Jazz / Charleston

- **1st Place**: 2012 Solo Jazz, *Sparx*, Cleveland, OH.
- **3rd Place**: 2012 Solo Charleston, *Stompology*, Rochester, NY.

C. Balboa

- **Finals**: 2014 Strictly Balboa w/Jennifer Lee, *International Lindy Hop Championships*, Washington, D.C.
- **3rd Place**: 2013 Amateur Strictly Balboa w/Annabel Truesdell Quisao, *All Balboa Weekend*, Independence, OH.
- **4th Place**: 2013 Amateur Jack & Jill, *All Balboa Weekend*, Independence, OH.
- **Finals**: 2012 Amateur Jack & Jill w/Nina Galicheva, *All Balboa Weekend*, Independence, OH.

D. Blues

- **Finals**: 2013 Solo Riffin' Competition, *Steel City Blues*, Pittsburgh, PA.
- **1st Place**: 2012 Solo Riffin' Competition, *Steel City Blues*, Pittsburgh, PA.
- **Finals**: 2010 Open Jack & Jill, *Steel City Blues*, Pittsburgh, PA.

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<!-- # References -->

<!-- \begin{longtable}{p{7cm} p{8.6cm}} -->
<!--   \textbf{Erica R.H. Fuchs} & \textbf{Jeremy J. Michalek} \\ -->
<!--   Professor & Professor \\ -->
<!--   Department of Engineering and Public Policy & Department of Engineering and Public Policy \\ -->
<!--   Carnegie Mellon University & Department of Mechanical Engineering \\ -->
<!--   Baker Hall 131E & Carnegie Mellon University \\ -->
<!--   5000 Forbes Avenue & Scaife Hall 324 \\ -->
<!--   Pittsburgh, PA 15213 & 5000 Forbes Avenue \\ -->
<!--   Phone: +1 (412) 268-1877 & Pittsburgh, PA 15213 \\ -->
<!--   Email: erhf@andrew.cmu.edu & Phone: +1 (412) 268-3765 \\ -->
<!--   & Email: jmichalek@cmu.edu \\ -->
<!--   &  \\ -->
<!--   \textbf{Elea McDonnell Feit} & \textbf{Valerie J. Karplus}\\ -->
<!--   Associate Professor of Marketing & Associate Professor\\ -->
<!--   LeBow College of Business & Department of Engineering and Public Policy\\ -->
<!--   Drexel University & Carnegie Mellon University\\ -->
<!--   Gerri C. LeBow Hall 828 & Scott Hall 5113\\ -->
<!--   3141 Chestnut Street & 5000 Forbes Avenue\\ -->
<!--   Philadelphia, PA 19104 & Pittsburgh, PA 15213\\ -->
<!--   Phone: +1 (215) 571-4054 & Email: vkarplus@andrew.cmu.edu \\ -->
<!--   Email: efeit@drexel.edu & \\ -->
<!--   &  \\ -->
<!--   \textbf{Peter Fox-Penner} & \textbf{Zoe Szajnfarber} \\ -->
<!--   Director, Institute for Sustainable Energy & Professor \& Department Chair\\ -->
<!--   Questrom School of Business & Engineering Management and Systems Engineering\\ -->
<!--   Boston University & George Washington University\\ -->
<!--   Rafik B. Hariri Building, Room 514A & Science \& Engineering Hall 2670\\ -->
<!--   595 Commonwealth Ave. & 800 22nd Street, NW\\ -->
<!--   Boston, MA 02215 & Washington, DC 20052 \\ -->
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<!--   Phone: +1 (617) 353-4298 & Phone: +1 (202) 994-7153\\ -->
<!--   Email: pfoxfp@bu.edu & Email: zszajnfa@gwu.edu\\ -->
<!-- \end{longtable} -->
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